Uinsure Backend Technical Assessment

As part of your application to Uinsure, we ask all candidates to complete a take home technical test which takes the form of a small application revolving around storing home insurance policies.

Background

Uinsure sells home insurance of which there are two types

1. Household
2. Buy to Let

A policy can have three actions which a consumer can carry out

1. Sell
2. Cancel
3. Renew

You can buy or renew a policy using the following payment methods

1. Card
2. Direct Debit
3. Cheque

Sell

When selling a policy

1. Policy Reference must be unique
2. A policy can only be sold up to 60 days in advance
3. A policy must always be 1 year in length
4. A policy must have at least 1 but no more than 3 policyholders
5. All policyholders must be over 16 years old on the start date of the policy

Renew

When renewing an existing policy the following criteria apply

1. A policy can only be renewed 30 days before the end date of the current policy
2. A policy cannot be renewed after the end date
3. When creating a renewal against a policy
    a. which is set to auto renew true a payment is created
    b. which is set to auto renew false a payment is not created

Cancel

When cancelling an existing policy the following criteria apply

1. The refund amount changes based on the cancellation date
    a. Cancellation before the policy start date results in a full refund
    b. Cancellation on or after the policy start date and within the 14 day cooling off period results in a full refund
    c. Cancellation after the 14 day cooling off period results in a pro rata refund based on the number of days of insurance not used
2. The refund can only be made to the same method as the original payment

Functional Requirements

The requirements are categorised using the MOSCOW prioritisation framework. A consumer of the Api should be able to achieve all of the must and at least 1 of the should have requirements. The could have requirements are optional.

Must

1. Sell a policy
2. Retrieve a Policy

Should

1. Cancel the policy
2. Renew a policy

Could

1. Calculate the cost to cancel a policy before the policy has actually been cancelled.
2. Not issue a refund if the policy has had a claim made against it
3. Prevent an auto renewal policy renewal being paid for by cheque

Constraints

The solution should

1. deliver a REST Api written in C# using the policy information defined in appendix 1
2. have automated tests
3. return informative responses to the consumer
4. be shared in a publicly accessible github repo

The goal of this test is to act as the focus of a 2 way technical discussion during the interview to understand the steps & considerations taken to deliver the solution.

We have intentionally kept the scope of this take home test open to interpretation as we want to let candidates demonstrate their skills. There are no wrong answers and we expect candidates may take different approaches and focus on different areas. We are not expecting production ready code or a full working solution but automated tests should be included to cover areas candidates feel are most appropriate.

Appendix - Policy Information

Home insurance policies require the following minimum information to be stored. Additional fields can be added to meet the requirements as needed.

Policy

- Unique Reference
- StartDate
- End Date
- Amount
- HasClaims
- AutoRenew

Policyholder

- First Name
- Last Name
- Date of Birth

Property (covered by the Insurance)

- Address Line 1 – Required
- Address Line 2
- Address Line 3
- Postcode - Required with max length of 8 characters

Payment

- PaymentReference
- Type
- Amount

Refund

- RefundReference
- Type
- Amount